



TECHNICAL PROJECT PRESENTATION

AI Resume Screening & Job Matching System

Prepared for portfolio & recruiter discussion

PYTHON

FLASK

NLP

SQLITE

Core project details reflect the implementation as designed; final specifics to be verified on submission.

What This Project Does

An AI-powered web application that compares a candidate's resume with a job description **to estimate fit and guide improvement.**



Goal

Identify relevant skills, estimate job fit, and provide actionable improvement suggestions.



Use Case

Built as a practical AI/ML portfolio project modeled on real recruitment workflows.



Resume

PDF upload



Job Description

Target role requirements



Match Score + Suggestions

Fit estimate & skill gaps to close

THE PROBLEM

Manual Resume Review Doesn't Scale

Recruiters often need to review many resumes against different job descriptions
— a process that is slow and inconsistent.



High Volume

Many resumes must be checked against every open role.



Time-Consuming

Manual, line-by-line comparison is slow at scale.

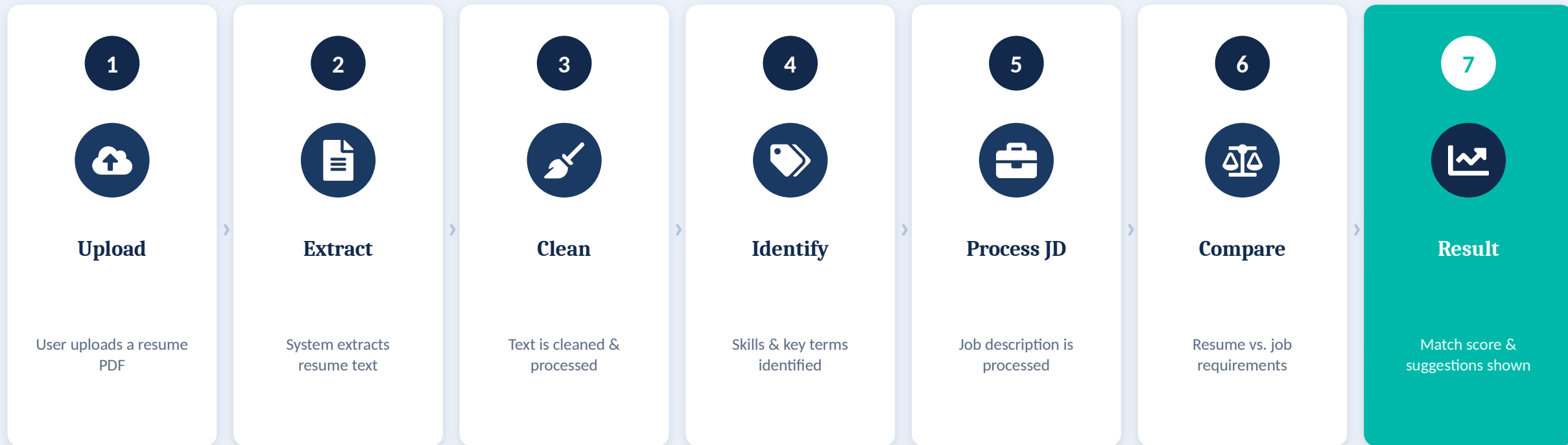


Missed Signals

Fatigue and inconsistency can cause relevant skills to be overlooked.

The system automates the first pass — while keeping every result understandable and explainable.

How It Works



SQLite can store users, resumes, job descriptions, and results throughout the pipeline.

System Architecture



SQLite persists users, resumes, job descriptions, and results across the flow.



Technology Stack



Python

Application & data-processing logic



Flask

Web backend / API layer



PyMuPDF

PDF text extraction



NLP Techniques

Resume & job analysis



SQLite

Lightweight data storage



HTML/CSS/JS

User interface

WALKTHROUGH

Example: Input → Output



Job Description

Python developer requiring:

Python • SQL • Flask • Machine Learning



Candidate Resume

Listed skills:

Python • SQL • Java • Machine Learning



RESULT



Matched Skills

Python, SQL, Machine Learning



Missing Skill

Flask

The exact scoring method should be documented according to the final implemented version.

Challenges & Learning



Inconsistent PDF Formatting

PDFs vary widely, which affects text-extraction quality and downstream accuracy.



Keyword Matching Limits

Keyword matching alone can produce false positives or miss related terminology.



Explainability Matters

Scoring needs to be explainable rather than acting as a black box.



Key Takeaway

This project demonstrates Python, backend development, text processing, and applied AI/ML thinking.

Future Improvements



Semantic Similarity

Use sentence embeddings for deeper meaning-based matching.



Better Extraction

Improve skill and entity extraction accuracy.



Resume Ranking

Rank multiple candidates for a single role.



Recruiter Dashboard

Add a dashboard with authentication for recruiters.



Job Recommendations

Suggest roles based on candidate profile.



Cloud Deployment

Deploy the system for broader, scalable access.